

Mersenne GRAMMAR — 5 k-Types Complete

Author: Martin Doina | $p_e = m^2 \times (p_a \times p_b) \pm k$ | Added: p_c^2

$p_e = m^2 \times (p_a \times p_b) \pm k$

diff: $p_c - p_d$ 2p+p: $2p_c + p_d$ 2p-p: $2p_c - p_d$ single: $\pm p_c$ square: $p_c^2 \pm p_d$

50/50

Found

0

Gaps

Done: 50/50

M#	p_e	m	p_a	p_b	Prod	m ² ×Prod	k	k-Type	✓
M3	5	1	2	2	4	4	1	2p-p	✓
M4	7	1	2	2	4	4	3	2p-p	✓
M5	13	1	2	2	4	4	9	2p+p	✓
M6	17	1	2	2	4	4	13	2p+p	✓
M7	19	1	2	2	4	4	15	2p+p	✓
M8	31	1	2	2	4	4	27	sq+	✓
M9	61	1	2	2	4	4	57	2p+p	✓
M10	89	1	2	5	10	10	79	2p+p	✓
M11	107	1	2	2	4	4	103	2p+p	✓
M12	127	1	2	2	4	4	123	2p+p	✓
M13	521	1	2	89	178	178	343	2p+p	✓
M14	607	1	3	31	93	93	514	diff	✓
M15	1,279	1	2	2	4	4	1,275	2p+p	✓
M16	2,203	1	5	127	635	635	1,568	sq+	✓
M17	2,281	1	3	607	1,821	1,821	460	diff	✓
M18	3,217	1	5	607	3,035	3,035	182	sq+	✓
M19	4,253	1	3	1,279	3,837	3,837	416	sq+	✓
M20	4,423	1	2	2	4	4	4,419	2p+p	✓
M21	9,689	1	2	19	38	38	9,651	2p+p	✓
M22	9,941	1	2	19	38	38	9,903	2p+p	✓
M23	11,213	1	13	127	1,651	1,651	9,562	diff	✓
M24	19,937	1	2	19	38	38	19,899	2p+p	✓
M25	21,701	1	3	607	1,821	1,821	19,880	2p-p	✓
M26	23,209	1	7	3,217	22,519	22,519	690	sq+	✓
M27	44,497	1	2	2,281	4,562	4,562	39,935	2p+p	✓
M28	86,243	1	19	4,253	80,807	80,807	5,436	diff	✓
M29	110,503	1	31	4,253	131,843	131,843	-21,340	sq-	✓
M30	132,049	1	107	521	55,747	55,747	76,302	diff	✓
M31	216,091	223	2	2	4	198,916	17,175	2p-p	✓
M32	756,839	409	2	2	4	669,124	87,715	2p-p	✓
M33	859,433	152	2	13	26	600,704	258,729	2p+p	✓
M34	1,257,787	551	2	2	4	1,214,404	43,383	2p-p	✓
M35	1,398,269	316	2	7	14	1,397,984	285	2p+p	✓
M36	2,976,221	853	2	2	4	2,910,436	65,785	2p-p	✓
M37	3,021,377	379	3	7	21	3,016,461	4,916	sq-	✓
M38	6,972,593	1078	2	3	6	6,972,504	89	2p-p	✓
M39	13,466,917	466	2	31	62	13,463,672	3,245	2p+p	✓
M40	20,996,011	1449	2	5	10	20,996,010	1	2p-p	✓
M41	24,036,583	59	3	2,281	6,843	23,820,483	216,100	sq+	✓
M42	25,964,951	11	3	21,701	65,103	7,877,463	18,087,488	sq-	✓
M43	30,402,457	2753	2	2	4	30,316,036	86,421	2p+p	✓

M#	p_e	m	p_a	p_b	Prod	m ² ×Prod	k	k-Type	✓
M44	32,582,657	41	2	9,689	19,378	32,574,418	8,239	2p-p	✓
M45	37,156,667	3051	2	2	4	37,234,404	-77,737	2p-p	✓
M46	42,643,801	3236	2	2	4	41,886,784	757,017	2p+p	✓
M47	43,112,609	2075	2	5	10	43,056,250	56,359	2p+p	✓
M48	57,885,161	2033	2	7	14	57,863,246	21,915	2p+p	✓
M49	74,207,281	1141	3	19	57	74,207,217	64	2p+p	✓
M50	77,232,917	2779	2	5	10	77,228,410	4,507	2p+p	✓
M51	82,589,933	76	3	4,423	13,269	76,641,744	5,948,189	2p-p	✓
M52	136,279,841	1899	2	19	38	137,035,638	-755,797	2p-p	✓

50/50 — GRAMMAR COMPLETE!
The Mersenne Ring is CLOSED with 5 k-types.